Examiner only

10. Halogenoalkanes are hydrolysed when heated with aqueous sodium hydroxide.

(a) (i) Write the equation to show the hydrolysis of 3-chloro-2-methylpentane with aqueous sodium hydroxide. You should show clearly the structure of the organic reagent and product. [1]

(ii) State the name of the mechanism for the reaction taking place in part (i). [1]

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(iii) Describe a chemical test to show that this hydrolysis reaction has occurred. Include the test and the result expected. [2]

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Examiner only

(b) The rate of reaction between a halogenoalkane and aqueous sodium hydroxide was measured when different concentrations of reagents were used. The results are shown in the table.

Experiment	Initial concentration OH^-(aq) / mol dm^{-3}	Initial concentration halogenoalkane / mol dm^{-3}	Initial rate of reaction / mol dm^{-3} s^{-1}
1	0.10	0.10	1.20×10^{-5}
2	0.10	0.20	2.40×10^{-5}
3	0.10	0.30	3.60×10^{-5}
4	0.20	0.10	1.20×10^{-5}
5	0.30	0.10	1.20×10^{-5}

(i) Use these data to deduce how the concentration of each reagent affects the rate of reaction. Explain how you reached your conclusions. [2]

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(ii) What would be the effect on the initial rate of reaction if the halogen in the halogenoalkane were changed from chlorine to bromine? You should assume that this is the only change made. Explain your answer. [3]

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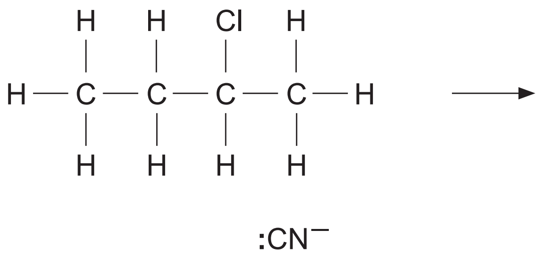
10. *(a)* Halogenoalkanes react with potassium cyanide in a similar way to their reaction with $OH^-(aq)$.

(i) Complete the mechanism of the reaction between 2-chlorobutane and potassium cyanide. You should include relevant dipoles and curly arrows to show the movement of electrons. [4]



(ii) Name the type of reaction mechanism shown in part (i). [1]

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(b) When 2-chlorobutane is heated with sodium hydroxide dissolved in ethanol three different organic products are formed.

(i) Draw the displayed formulae of the **three** products. Name each product. [3]

(ii) Name the type of reaction taking place. [1]

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Examiner only

(c) Until recently chlorofluorocarbons, CFCs, were used in many commercial products but nowadays hydrofluorocarbons, HFCs, are preferred.

Explain why HFCs are preferred to CFCs. You do **not** need to include the mechanism or equation for any reaction which you describe. [5]

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14

10. *(a)* Halogenoalkanes can be hydrolysed using water in a similar way to using aqueous sodium hydroxide.

A student carried out an experiment to investigate the rate of reaction for the hydrolysis of halogenoalkanes using water. The student used aqueous ethanol to dissolve the halogenoalkane and then added a few drops of aqueous silver nitrate. He timed how long it took to produce a precipitate. He obtained the results shown in the table.

Halogenoalkane	Time / s
1-chloropropane, C_3H_7Cl	300
1-bromopropane, C_3H_7Br	90
1-iodopropane, C_3H_7I	15

The student tried to explain these results and he looked on the internet to find the following data.

Bond	Bond enthalpy / kJ mol^{-1}
C—H	413
C—C	348
C—F	485
C—Cl	328
C—Br	276
C—I	240

Element	Electronegativity
chlorine	3.16
bromine	2.96
iodine	2.66
carbon	2.55

Examiner only

(i) Use both sets of data to explain the student's results. Include an equation to show the hydrolysis reaction between a halogenoalkane and aqueous sodium hydroxide and name this type of reaction mechanism. [6 QER]

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(ii) Write an **ionic** equation, including state symbols, for a reaction that produces a silver halide precipitate. [1]

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(iii) Suggest a practical method by which the student could have obtained these results. [2]

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(iv) Suggest the difference that the student would have observed in his experiments if he had not added ethanol to the water before adding the halogenoalkane. [1]

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Examiner only

(b) Chlorofluorocarbons, CFCs, were historically used for a variety of commercial and domestic purposes but nowadays their use is very restricted.

(i) Outline the adverse environmental effects of the use of CFCs.

You do not need to include any equations for the reactions involved. [4]

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(ii) Use relevant data in part *(a)* to explain why hydrofluorocarbons, HFCs, have replaced CFCs in many of their uses. [2]

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16

10. *(a)* Students were discussing the reactivity of organic compounds. One said that reactivity was due to dipoles produced by differences in electronegativity.

Show that this is not correct by discussing the difference in reactivity between alkanes and alkenes. You should include reference to the bonding in both series.

[6 QER]

Examiner only

(b) In the 19th century Cannizzaro discovered a reaction that involved disproportionation. Such reactions involve the same substance being both oxidised and reduced.

The equation for the Cannizzaro reaction is shown.

$$2C_6H_5CHO \ + \ H_2O \longrightarrow C_6H_5CH_2OH \ + \ C_6H_5COOH$$

(i) State why this reaction is classified as *disproportionation*. [1]

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(ii) A chemist used 9.50 g of C_6H_5CHO in the reaction above and made 3.39 g of $C_6H_5CH_2OH$.

Calculate the percentage yield that this mass of $C_6H_5CH_2OH$ represents. [3]

Percentage yield = ... %

(iii) Cannizzaro's reaction is usually carried out using aqueous sodium hydroxide, rather than with water as shown in the equation above.

Write the equation for the Cannizzaro reaction of C_6H_5CHO with aqueous sodium hydroxide. [1]

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10. The crystallisation of sodium ethanoate from a super-saturated solution is used to release heat in reusable hand warmers.

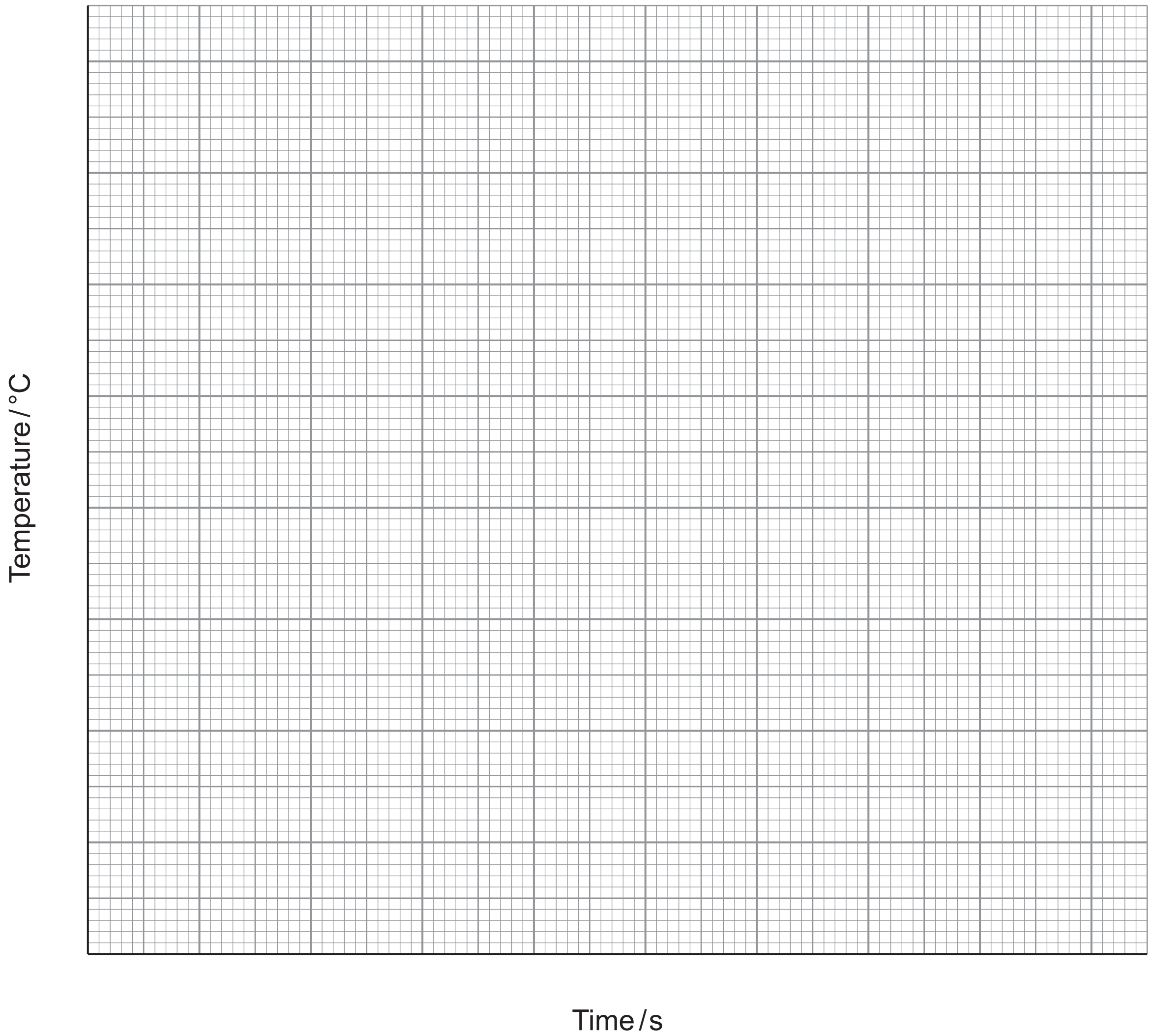
(a) A super-saturated solution of sodium ethanoate was made by dissolving 320 g of hydrated sodium ethanoate ($CH_3COONa.3H_2O$) in 60 cm^3 of hot water. It was then allowed to cool to room temperature, which was measured as 17 °C.

A thermometer was added to the solution, which caused the sodium ethanoate to start crystallising. The temperature of the process was recorded every 30 seconds for 3 minutes. The results are shown below:

Time / s	Temperature / °C
0	17
30	27
60	35
90	41
120	40
150	39
180	38

Examiner only

(i) Plot the results on the graph paper below. [2]



(ii) Use your graph to calculate the maximum temperature change for this crystallisation. [2]

maximum temperature change = ... °C

Examiner only

(iii) Use the **total mass** of the sodium ethanoate solution and the temperature change from the graph to calculate the enthalpy change of crystallisation per mole of sodium ethanoate. Assume the density of water is 1.00 g cm^{-3} and the specific heat capacity of sodium ethanoate solution is 4.18 J K^{-1} g^{-1}.

$M_r(CH_3COONa.3H_2O) = 136$ [4]

enthalpy change = ... kJ mol^{-1}

(iv) Suggest a reason why the experimental enthalpy change is often lower than the theoretical enthalpy change. [1]

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(b) Sodium ethanoate can be made in a neutralisation reaction. Complete the following equation: [2]

$$\ldots\ldots\ldots\ldots \quad + \quad \ldots\ldots\ldots\ldots \longrightarrow CH_3COONa \; + \; CO_2 \; + \; H_2O$$

(c) The carboxylic acid used to produce sodium ethanoate can be produced using an oxidation reaction.

(i) Name the reagents and give the expected observations. [2]

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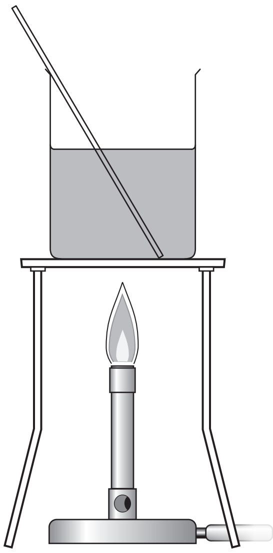
(ii) A student proposed that the apparatus below should be used to perform this oxidation reduction experiment.



The teacher said that this would not work and would be unsafe. Draw a labelled diagram of the apparatus that should be used in this experiment. [3]

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END OF PAPER

Surname	Centre Number	Candidate Number
First name(s)		2

GCE AS/A LEVEL

2410U20-1

S23-2410U20-1

THURSDAY, 25 MAY 2023 – MORNING

CHEMISTRY – AS unit 2
Energy, Rate and Chemistry of Carbon Compounds

1 hour 30 minutes

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1. to 7.	10	
Section B	8.	6	
	9.	6	
	10.	18	
	11.	10	
	12.	13	
	13.	17	
	Total	80	

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ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:
- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.
Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.8**.

10. (a) The enthalpy of neutralisation of an acid is defined as the enthalpy change when 1 mol of aqueous H^+ ions is neutralised by aqueous OH^- ions according to the equation shown. The reaction is exothermic.

$$H^+(aq) \quad + \quad OH^-(aq) \longrightarrow H_2O(l)$$

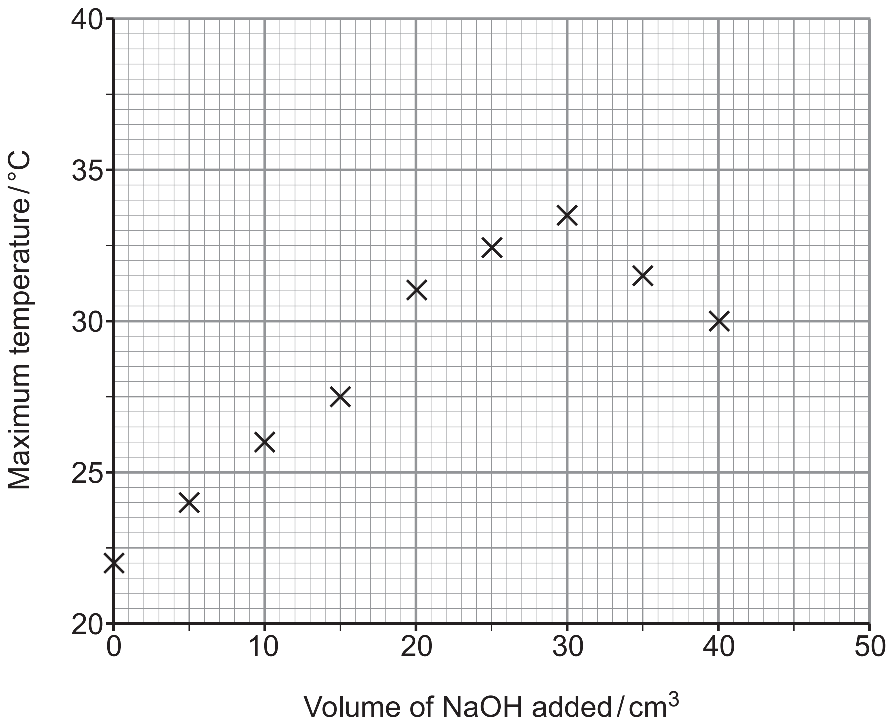
Some students followed the instructions below to determine the enthalpy change of neutralisation of methanoic acid, HCOOH.

1. Weigh 24.7 g of methanoic acid and mix with water to make 250 cm^3 of solution. Record the temperature of the solution.
2. Transfer eight 25.0 cm^3 portions of this solution into eight insulated cups.
3. Using a burette add 5.0 cm^3 of aqueous sodium hydroxide to the solution in the first cup. Stir and record the maximum temperature reached.
4. Add the following volumes of aqueous sodium hydroxide to each of the remaining cups in turn:

   10.0 cm^3 15.0 cm^3 20.0 cm^3 25.0 cm^3 30.0 cm^3 35.0 cm^3 40.0 cm^3

   Stir and record the maximum temperature reached in each cup.
5. Plot a graph of maximum temperature reached against volume of sodium hydroxide added.

Their results are plotted in the graph.



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(i) Name the apparatus used to transfer exactly 25.0 cm^3 of methanoic acid solution into the insulated cups. [1]

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(ii) State why a higher maximum temperature is recorded when increasing volumes of sodium hydroxide are added. [1]

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(iii) Explain why the maximum temperature recorded decreases when more than 30 cm^3 of sodium hydroxide is added. [2]

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(iv) On the graph, draw **one** straight line through the points that show an increase in maximum temperature and **another** straight line through the points that show a decrease in maximum temperature. [1]

(v) From the graph, deduce the volume of sodium hydroxide needed to neutralise 25.0 cm^3 of the methanoic acid solution and the temperature increase at that point.

Assume that the initial temperature of every 25.0 cm^3 of methanoic acid solution is 22.0 °C. [2]

Volume = ... cm^3

Temperature increase = ... °C

(vi) Use your answers to part (v) to calculate the amount of heat released by the neutralisation reaction. [2]

Heat released = ... J

(vii) Calculate the number of moles of methanoic acid in 25.0 cm^3 of the solution and hence the enthalpy change of neutralisation of methanoic acid. [3]

Enthalpy change of neutralisation = ... $kJ\,mol^{-1}$

Examiner only

(b) The experiment is repeated using hydrochloric acid instead of methanoic acid and a more negative value of the enthalpy change of neutralisation is calculated.

Suggest and explain a reason for this difference. [2]

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(c) (i) Write the equation for the reaction that occurs when solid copper(II) carbonate is added to aqueous methanoic acid to form aqueous copper(II) methanoate, $(HCOO)_2Cu$.

Include state symbols. [2]

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(ii) State what is observed during the reaction in part (i). [2]

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